MRC GHANA 2027
COACH PREPARATION GUIDE

Coach Preparation Guide

Planning, building and team management

29–30 January 2027 · Accra, Ghana · Venue to be announced

The current official H.E.R.O. rules take precedence
This guide is a plain-English summary prepared by the MRC Ghana 2027 organisers. Always check each sport's official rules, linked from https://www.mrcghana.com/categories.

Your role as a coach

- You must be at least 20 years old. Bring one assistant for every three teams you register.
- Each team has 2 to 3 members (in most sports), all at least 10 years old.
- Each team names one Technician, the only team member allowed in the waiting and competition area. The Technician may change between attempts or matches.
- One robot per team. Robots may not be swapped during the competition or shared between teams.
- Rule ZERO applies: if you are not sure something is allowed, assume it is not and ask the organisers.

Suggested preparation plan

A suggested plan to adapt to your school or club calendar. Only the registration deadline and competition dates are fixed.

When	What to do
Now	Choose sports, check every student's age band, read the official rules and form teams. Check equipment lists before buying parts or drones.
By 15 January 2027	Register every team. Collect guardian details and consent for every student under 18.
Build season	Build and test against the size, weight and equipment rules. Practise runs and matches, and practise explaining how the robot works.
Final week	Work through the Technical Inspection Checklist, charge batteries, pack spares and protective eyewear, and check organiser updates on reporting times.
29–30 January 2027	Report for technical inspection, then compete.
Afterwards	Follow results and any international selection announcements from the organisers.

What to have ready for registration

- Organisation details (school, club or institution)
- Coach details and team members, including dates of birth
- Guardian details and consent for every minor
- Robot details and emergency contacts

Preparation checklists by sport

Drones

- [ ] Confirm your drone model meets the size, weight and battery rules (or request pre-approval).
- [ ] Fit full-perimeter propeller guards and remove any sharp parts.
- [ ] Test altitude hold at 1–2 m and practise using the kill switch.
- [ ] Prepare two charged batteries and a safe charging/storage bag.
- [ ] Practise hoop runs, precision landings and fast route planning.

- [ ] Pack protective goggles for the pilot.

Drone Soccer 3×3

- [ ] Confirm with the organisers which drone-ball model is specified for MRC Ghana 2027.
- [ ] Form an alliance of three teams from your school or institution.
- [ ] Agree alliance roles (Defender, Attacker, Golden Ball Drone) and practise together.
- [ ] Check LED colours, weight and size; prepare spare batteries.
- [ ] Practise penalty attempts and returning to your half after goals.

Marathon

- [ ] Choose Basic or Advanced and check your kit against that division's rules.
- [ ] Confirm size (25 × 25 cm), weight (up to 1 kg) and height (5–15 cm).
- [ ] Fit and test the start/stop button.
- [ ] Practise on unfamiliar tracks with curves and intersections (go straight at crossings).
- [ ] Print the official practice track sample and time your runs.
- [ ] Bring a laptop with your program ready to explain to judges.

Football 3×3

- [ ] Form an alliance of three teams from your school or institution.
- [ ] Check robot size, weight (700–800 g) and height; test the kicker at full extension.
- [ ] Practise driving, passing and shooting with a 52 mm LEGO ball.
- [ ] Prepare spare parts and batteries for quick repairs between matches.
- [ ] Agree your alliance identifier (flag or colours).

Shooting

- [ ] Read the official design and shape rules before you start building.
- [ ] Check size (25 × 25 cm) and mass (up to 500 g); use only permitted materials.
- [ ] Program the one-second warning beep and single-dart firing.
- [ ] Practise at 120 cm and 160 cm and record your accuracy.
- [ ] Pack safety glasses and protective equipment.
- [ ] Bring a laptop with your program ready to explain to judges.

Archery

- [ ] Design a vertical bow mechanism and check size, weight and drawn length.
- [ ] Make and weigh several blunt-tipped arrows (under 30 g).
- [ ] Program the five-second delayed start and automatic height adjustment.
- [ ] Print the official target and practise at all four distances.
- [ ] Pack safety glasses and protective equipment.

- [ ] Bring a laptop with your program ready to explain to judges.

Exhibition

- [ ] Register your interest and follow this page for updates.
- [ ] Start documenting your project idea, design process and testing.
- [ ] Check the official H.E.R.O. page regularly for the published rules.

On competition day

- Be on time for technical inspection. In some sports, missing the inspection slot or failing inspection means exclusion or disqualification.
- Protective eyewear: the Drones pilot wears goggles; Shooting and Archery teams need safety glasses and protective equipment.
- Reporting times and the programme for both days will be published by the organisers on the Programme page.

Useful links

Sports, rules and age bands: https://www.mrcghana.com/categories
Category information sheets: https://www.mrcghana.com/resources
Register a team: https://www.mrcghana.com/register
Frequently asked questions: https://www.mrcghana.com/faq

Contact the organisers

- Email: partnerships@makersplacegh.com
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- Website: www.mrcghana.com